

Abhishek Singh

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Work Experience

Motilal Nehru National Institute of Technology(MNNIT), Allahabad

Junior Research Fellow Dec, 2025 – Present

- Researching advanced deep learning models like Transformers, ViTs, CNNs and GANs for medical imaging.
- Designing AI models for a Portable 3D X-ray Imaging Diagnostic System aimed at early detection of TB and COPD.

Indian Institute of Technology(IIT), Roorkee

Junior Research Fellow Aug, 2025 – Oct, 2025

- Worked on a DRDO-funded project titled “Technique for Context-Based Security Analysis of Platform.”
- Conducted literature survey on existing cryptographic algorithms used in secure platforms.
- Identified limitations of conventional approaches and emphasized the need for lightweight cryptographic algorithms for resource-constrained systems.

Education

Indian Institute of Information Technology, Lucknow

July, 2023 – Jun, 2025
M.Tech in Computer Science & Engineering (Data Science)

CGPA: 7.93/10

Relevant Coursework: Big Data, Deep Learning, Natural Language Processing, Machine Learning

Manipal University Jaipur

July, 2018 – Jun, 2022
B.Tech in Computer Science & Engineering

CGPA: 8.10/10

Relevant Coursework: Operating Systems, Computer Networks, Data Structures and Algorithms, Database Management Systems, Object-Oriented Programming

Skills

Languages/Tools: Python, C/C++, SQL, Jupyter Notebook, Kaggle Notebook

LLMs/GenAI: Generative Adversarial Networks (GANs), LangChain, Retrieval Augmented Generation (RAG)

ML/DL: CNN, RNN, LSTM, GRU, Transformers, Transfer Learning

Libraries/Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, Pandas, NumPy, Hadoop

Projects

ZS-OffDR: Zero-Shot Offensive Meme Detection and Reasoning

Python, PyTorch, VLMs

- Built **ZS-OffDR**, a multitask framework for classifying offensive memes and generating zero-shot reasoning using image and OCR-extracted Hindi text from **7,329 memes**.
- Achieved **+8.4% F1-score** on Level-1 (Offensive/Non-Offensive) and **+5.3%** on Level-2 (Implicit/Explicit) classification over strong baselines.
- Used **Qwen2.5-VL**, **Gemini Pro**, **DeepSeek-VL**, and **LLaVA-1.5** to generate natural language reasoning without additional training.
- Benchmarked explanation quality using **perplexity scores** and **semantic similarity** with **250 human-written explanations**; reduced inference time by **32%** through batch optimization.

CliniQ: MedQA using DeepSeek-8B

Python, LoRA, Unsloth, Hugging Face

- Developed an AI-driven medical question-answering system leveraging DeepSeek-8B for accurate, context-aware healthcare responses.
- Fine-tuned the model using LoRA, achieving **~50%** model compression without compromising accuracy.
- Integrated **Chain-of-Thought prompting** to significantly improve answer reasoning and QA performance.
- Accelerated training by **2×** using Unsloth optimizations and tracked progress with Weights & Biases (W&B).

AskTube: YouTube QA Chatbot

LangChain, Streamlit, FAISS, GenAI

- Built a transcript-driven RAG chatbot with FAISS for semantic retrieval and GenAI for context-aware responses.
- Integrated LangChain pipelines with optimized chunking, embeddings, and prompt engineering to improve accuracy and reduce hallucinations.
- Deployed an interactive Streamlit UI, enabling natural language video exploration and enhancing user engagement.

Certifications

Deep Learning - GUVI, IIT Madras

Machine Learning - Verzeo